

Identifying Conics and Reading Their Key Features

Answer Key

Algebra 2

Quick Answers

- | | |
|---|---|
| 1. 7 | 7. (a) the x-coordinate = -4 , (b) the |
| 2. (a) = 36, — | y-coordinate = -5 |
| 3. (a) the centre = $(5, 5)$, (b) the radius = 5 | 8. (a) the vertex distance = 6, (b) the |
| 4. (a) the x-coordinate = 3, (b) the | asymptote slope = $\frac{3}{4}$ |
| y-coordinate = -4 | 9. (a) the centre x = 5, (a) the centre y = -3 , |
| 5. (a) = -4 , — | (b) the radius = 4 |
| 6. (a) the semi-major axis = 5, (b) the focal | 10. (a) = $x = 3$, $y = 4$, $x = -4$, $y = -3$, — |
| distance = 4 | |
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What is verified

17 of 20 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 2, 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.C.7 – *problem 10*

HSG-GPE.A.1–A.3 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9*

Difficulty 1–5 of 5 across 20 tagged checks.

1. Radius of the circle $x^2 + y^2 = 49$.

Solution: In the standard form $(x - h)^2 + (y - k)^2 = r^2$, the right-hand side is r^2 , not r — so the radius is the square root of 49. 7

2. $9x^2 + 4y^2 = 36$. (a) Product of the squared-term coefficients. (b) Name the conic and justify.

Solution (a): The coefficients are 9 and 4, so the product is 36

Solution (b) — instructor-judged. It is an *ellipse*. The product is positive, so both squared terms carry the same sign and the curve closes up; but $9 \neq 4$, so the two axes have different lengths and it is not a circle. (In standard form, $\frac{x^2}{4} + \frac{y^2}{9} = 1$.) Full credit names the conic *and* gives the same-sign, unequal-coefficient reason. A bare name, or the positive product without noticing the coefficients differ, is partial credit.

3. Circle with diameter endpoints $(2, 1)$ and $(8, 9)$. (a) Centre. (b) Radius.

Solution (a): The centre is the midpoint of any diameter.

$$\left(\frac{2+8}{2}, \frac{1+9}{2}\right) = \left(\frac{10}{2}, \frac{10}{2}\right)$$

(5, 5)

Solution (b): The radius is half the diameter, so halve the distance between the endpoints.

$$\text{diameter} = \sqrt{(8-2)^2 + (9-1)^2} = \sqrt{36 + 64} = 10$$

Half of that is

5

4. Vertex of $y = x^2 - 6x + 5$.

Solution (a): For $y = ax^2 + bx + c$ the vertex sits on the axis of symmetry $x = -b/(2a)$, with $a = 1$ and $b = -6$:

$$x = \frac{6}{2}$$

3

Solution (b): Substitute that x back into the equation.

$$y = 3^2 - 6(3) + 5 = 9 - 18 + 5$$

-4

So the vertex is $(3, -4)$ and the vertex form is $y = (x - 3)^2 - 4$.

5. $x^2 - 4y^2 = 16$. (a) Product of the squared-term coefficients. (b) Name the conic and justify.

Solution (a): The coefficients are 1 and -4 , so the product is

-4

Solution (b) — instructor-judged. It is a *hyperbola*. The product is negative, so the two squared terms carry opposite signs, and that is exactly the test that separates a hyperbola from a circle or an ellipse: the curve cannot close up, it opens into two branches. (In standard form, $\frac{x^2}{16} - \frac{y^2}{4} = 1$.) Full credit names the conic *and* ties it to the opposite signs; naming it from a remembered picture, with no reference to the equation, is partial credit.

6. Ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$. (a) Semi-major axis. (b) Focal distance.

Solution (a): The denominators are a^2 and b^2 ; the larger one, 25, sits under x^2 , so the long axis is horizontal and its half-length is $\sqrt{25}$.

5

Solution (b): For an ellipse the focal distance satisfies $c^2 = a^2 - b^2$ — subtract, because the foci sit *inside* the curve.

$$c = \sqrt{25 - 9} = \sqrt{16}$$

4

So the foci are at $(\pm 4, 0)$, inside the vertices at $(\pm 5, 0)$, as they must be.

7. Vertex of $y = x^2 + 8x + 11$.

Solution (a): With $a = 1$ and $b = 8$:

$$x = \frac{-8}{2}$$

-4

Solution (b): Substituting back,

$$y = (-4)^2 + 8(-4) + 11 = 16 - 32 + 11$$

-5

Vertex form: $y = (x + 4)^2 - 5$.

8. Hyperbola $\frac{y^2}{36} - \frac{x^2}{64} = 1$. (a) Vertex distance. (b) Asymptote slope.

Solution (a): The y^2 term is the positive one, so the branches open up and down and the vertices lie $\sqrt{36}$ above and below the centre.

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Solution (b): For this orientation the asymptotes are $y = \pm \frac{a}{b}x$ with $a = \sqrt{36} = 6$ and $b = \sqrt{64} = 8$.

$$\frac{6}{8}$$

The rising asymptote has slope

$\frac{3}{4}$

9. $x^2 + y^2 - 10x + 6y + 18 = 0$. (a) Centre. (b) Radius.

Solution (a): Complete the square in each variable. Half of -10 is -5 and half of 6 is 3 :

$$\begin{aligned}(x^2 - 10x + 25) + (y^2 + 6y + 9) &= -18 + 25 + 9 \\ (x - 5)^2 + (y + 3)^2 &= 16\end{aligned}$$

Reading $(x - h)^2 + (y - k)^2 = r^2$ gives $h = 5$ and $k = -3$: the centre is and

$x = 5$

$y = -3$

Solution (b): The right-hand side is $r^2 = 16$, so the radius is

4

10. Where $y = x + 1$ meets $x^2 + y^2 = 25$. (a) The points. (b) Secant, tangent or miss.

Solution (a): Substitute the line into the circle.

$$\begin{aligned}x^2 + (x + 1)^2 &= 25 \\ 2x^2 + 2x + 1 &= 25 \\ x^2 + x - 12 &= 0 \\ (x - 3)(x + 4) &= 0\end{aligned}$$

At $x = 3, y = 4$; at $x = -4, y = -3$. Both points are 5 from the origin, as the circle demands. The crossings are

and

$x = 3, y = 4$

$x = -4, y = -3$

.

Solution (b) — instructor-judged. The substitution always produces a quadratic, and the count of its real solutions is the picture: two distinct solutions means two crossing points, so the line is a secant — which is this case; one repeated solution means the line touches at exactly one point, a tangent; no real solution means the line misses the circle. Full credit links the count of real solutions to all three pictures, not just to the two points found here.
